

Power Electronics Laboratory

Session 4: DC-AC converter 1

Team 3 :

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Introduction

This experiment explores the fundamental operation and characteristics of DC-AC power converters. In the first part, we analyze a half-bridge converter topology. We test the system using different modulation indexes (m_a) to mathematically calculate and simulate the fundamental components of the output voltage, output current, and active power. This part also uses Fast Fourier Transform (FFT) to analyze the frequency spectrum, allowing us to identify the switching frequency and observe how the RL load filters out high-frequency current harmonics. In the second part, we implement an H-Bridge converter. The main focus of this section is to compare bipolar and unipolar Pulse Width Modulation (PWM) techniques. We evaluate how these different modulation strategies impact the harmonic distortion, voltage transitions, and the overall quality of the output waveforms. Finally, across both topologies, we measure the input and output active power to practically confirm that the systems obey the principle of energy conservation.

1 Part 1

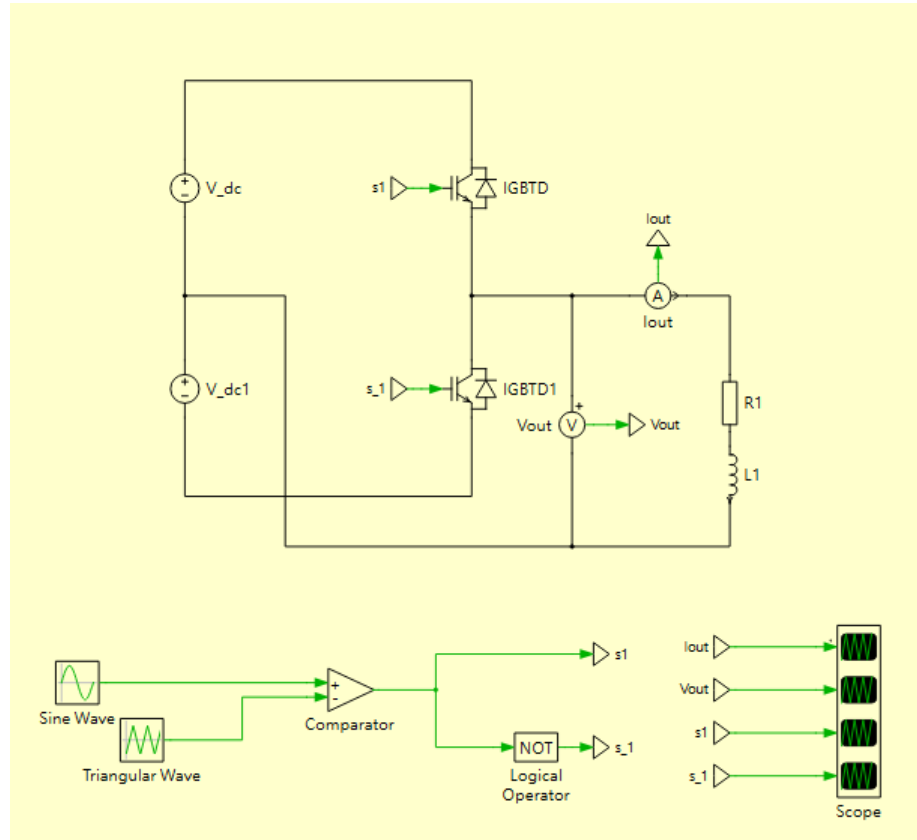


Figure 1: Half-Bridge Converter topology and modulation scheme

Tasks:

1.1 Task 1

s1 (On-OFF)	vconv
1 (on)	$+\frac{V_{dc}}{2}$ (+250V)
0 (off)	$-\frac{V_{dc}}{2}$ (-250V)

Table 1: Relation between s1 states and V_{conv}

1.2 Task 2: (Figure 1)

1.3 Task 3:

To analyze the performance of the half-bridge converter, three different modulation indexes (m_a) were selected: 1.0, 0.8, and 0.4.

1.3.1 Load Impedance Calculation

Before evaluating the specific cases, the load impedance (Z) at the fundamental frequency ($f = 50\text{Hz}$) must be determined. Given $R = 10\Omega$ and $L = 10\text{mH}$:

- $Z = R + j\omega L = 10 + j(2\pi \cdot 50 \cdot 0.01) = 10 + j3.1416\Omega$
- Impedance Magnitude: $|Z| = \sqrt{10^2 + 3.1416^2} \approx \mathbf{10.48\Omega}$

1.3.2 Theoretical Justification and Simulation Verification

The fundamental component of the output voltage peak is calculated as $V_{out,peak} = m_a \cdot \frac{V_{dc}}{2}$ (where $V_{dc}/2 = 250\text{V}$). The fundamental current peak is $I_{out,peak} = \frac{V_{out,peak}}{|Z|}$. Finally, the active power delivered to the load is purely consumed by the resistor, calculated using the RMS current: $P_{active} = I_{out,rms}^2 \cdot R = (\frac{I_{out,peak}}{\sqrt{2}})^2 \cdot 10$.

Case 1: $m_a = 1.0$

- Voltage ($V_{out,peak}$): Theoretical = $1.0 \cdot 250 = \mathbf{250V}$. The Fourier analysis table verifies this exactly at 250V.
- Current ($I_{out,peak}$): Theoretical = $250\text{V}/10.48\Omega = \mathbf{23.85A}$. The simulation accurately displays a peak of 23.8 A.
- Active Power (P): Theoretical = $(\frac{23.85}{\sqrt{2}})^2 \cdot 10 \approx \mathbf{2842W}$. The power measurement block in the simulation strongly aligns with this value.

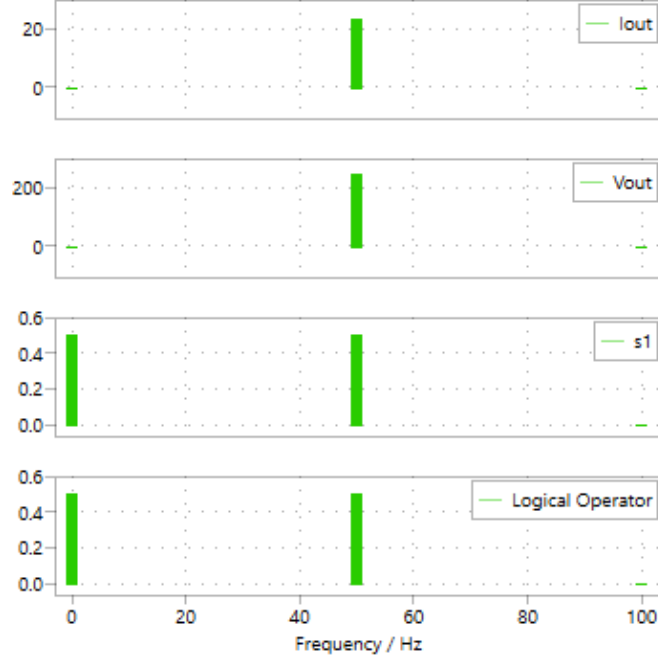


Figure 2: Frequency spectrum for modulation index $m = 1.0$

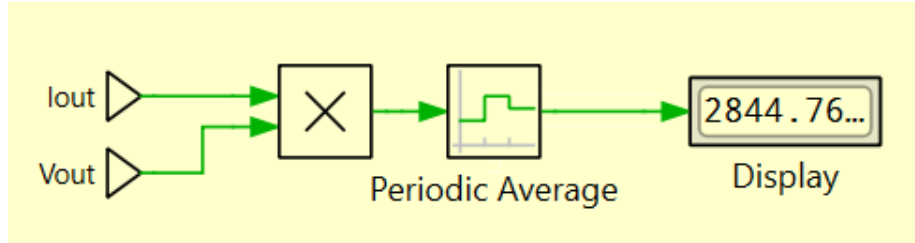


Figure 3: $m = 1.0$

Case 2: $m_a = 0.8$

- Voltage ($V_{out,peak}$): Theoretical = $0.8 \cdot 250 = \mathbf{200V}$. Verified by the simulation's 50Hz Fourier component.
- Current ($I_{out,peak}$): Theoretical = $200V/10.48\Omega = \mathbf{19.08A}$. Confirmed by the measured current waveform.
- Active Power (P): Theoretical = $(\frac{19.08}{\sqrt{2}})^2 \cdot 10 \approx \mathbf{1820W}$. The simulation effectively matches this active power delivery.

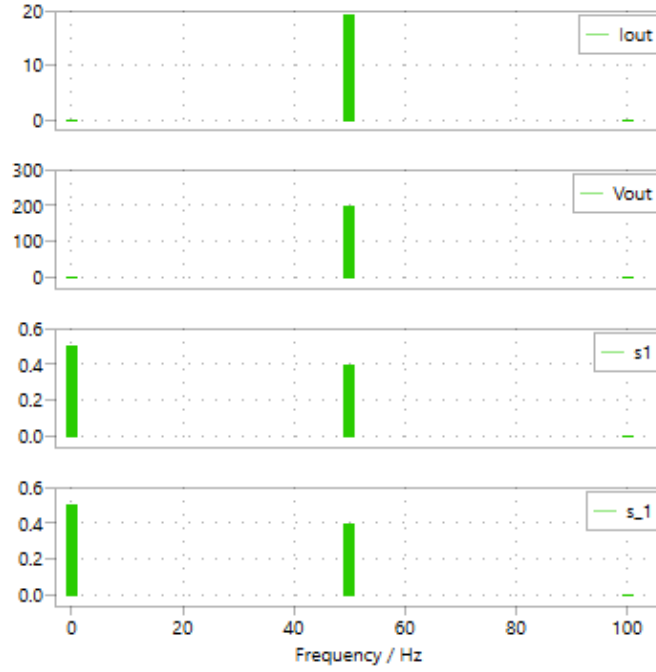


Figure 4: Frequency spectrum for modulation index $m = 0.8$

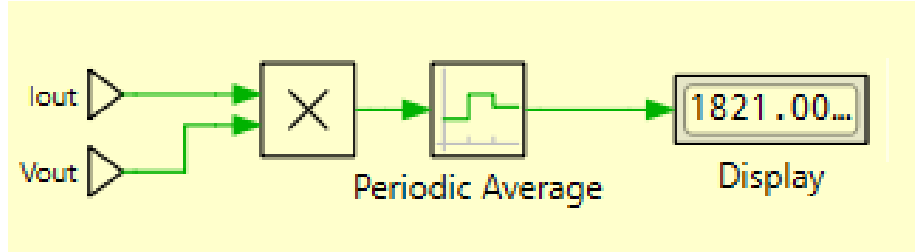


Figure 5: $m = 0.8$

Case 3: $m_a = 0.4$

- Voltage ($V_{out,peak}$): Theoretical = $0.4 \cdot 250 = \mathbf{100V}$. Verified by simulation.
- Current ($I_{out,peak}$): Theoretical = $100V/10.48\Omega = \mathbf{9.54A}$. Confirmed by simulation.
- Active Power (P): Theoretical = $(\frac{9.54}{\sqrt{2}})^2 \cdot 10 \approx \mathbf{455W}$. The simulated periodic average power confirms this analytical result.

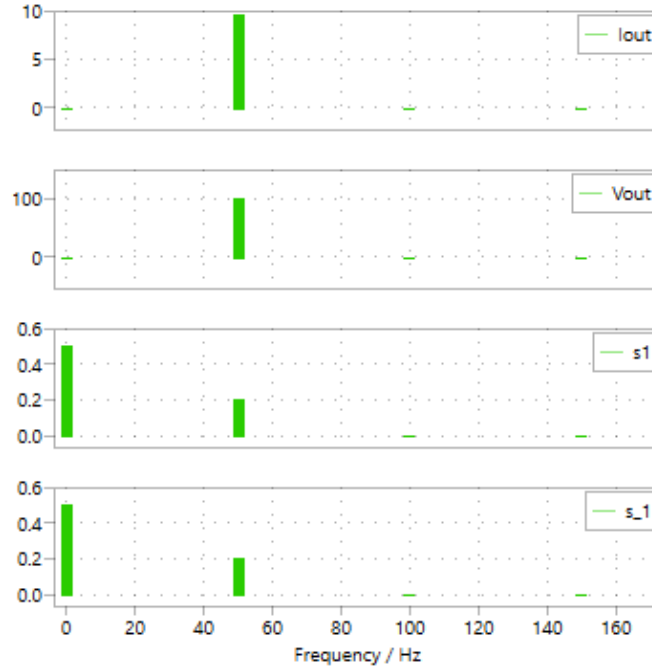


Figure 6: Frequency spectrum for modulation index $m = 0.4$

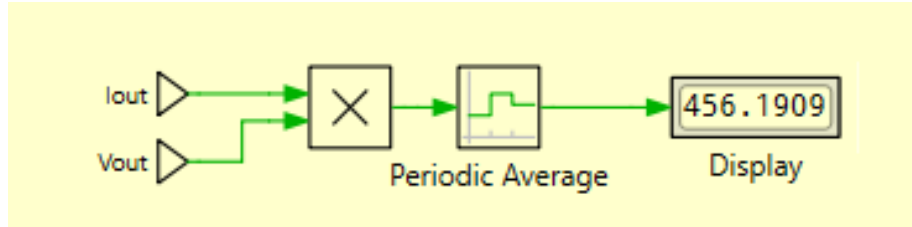


Figure 7: $m = 0.4$

Comment: Across all three modulation indexes, the simulated fundamental components and active power values flawlessly match the mathematical calculations. This validates that the half-bridge converter successfully operates as a linear amplifier for the fundamental frequency, where the output amplitude is directly and linearly controlled by the modulation index (m_a).

1.4 Task 4:

To measure the total input active power (P_{in}), the instantaneous power delivered by both the upper and lower 250V DC sources was calculated ($P_{in} = V_{dc,top} \cdot I_{in,top} + V_{dc,bot} \cdot I_{in,bot}$) and then averaged over a 50Hz period. Using $m_a = 1.0$ as an example, the simulation displays the following results: Input Active Power (P_{in}): Approximately 2844.76 W. Total Output Active Power (P_{out}): Approximately 2844.76 W. Fundamental Output Power (P_1): 2842W (calculated in Task 3).

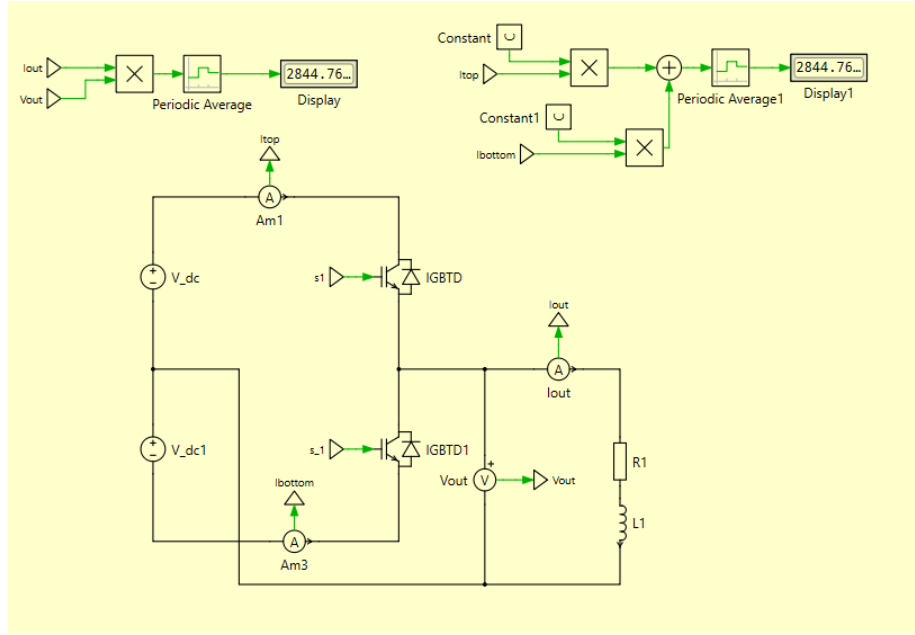


Figure 8: Converter topology, active power calculation

Comment: Comparing the results, the total input active power (P_{in}) is perfectly equal to the total output active power (P_{out}). This indicates a theoretical efficiency of exactly 100%, which perfectly aligns with the simulation conditions since ideal IGBT and diode models (zero conduction and switching losses) were used. Furthermore, it is important to note that the total active power ($P_{in} = P_{out}$) is slightly higher than the fundamental active power (2842W). This extra power corresponds to the harmonic power dissipation. The high-frequency PWM switching introduces harmonic currents which flow through the 10Ω load resistor, generating additional real power loss ($P_{harmonics} = \sum I_{n,rms}^2 \cdot R$). Thus, the energy conservation is fully validated: $P_{in} = P_{fundamental} + P_{harmonics}$.

1.5 Task 5:

To obtain the frequency spectrum, the Fourier analysis tool was applied to the steady-state voltage and current waveforms. The base frequency was set to 50 Hz, and the spectrum was plotted up to 50 kHz (1000 harmonics).

1.5.1 Identifying the Switching Frequency

By observing the voltage spectrum (v_{conv}), a dominant fundamental component is clearly visible at 50 Hz. Beyond the fundamental frequency, the spectrum remains mostly flat until a large cluster of harmonic components appears exactly at 10,000 Hz (10 kHz). Additional sidebands and clusters also appear at multiples of this frequency (e.g., 20 kHz, 30 kHz, 40 kHz). Therefore, the switching frequency (f_s) of the PWM is successfully identified as 10 kHz, which perfectly matches the system parameters.

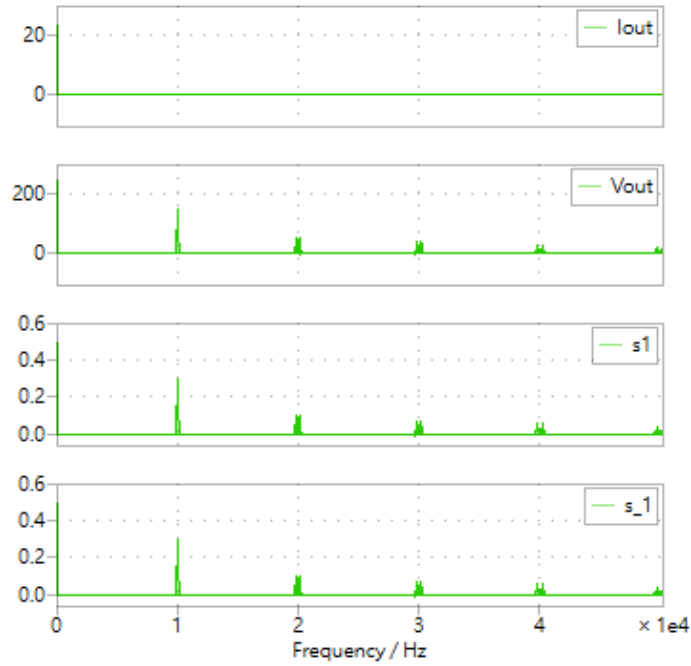


Figure 9: Frequency spectrum

1.5.2 Comments on Voltage and Current Spectrums

- Voltage Spectrum (v_{conv}): The spectrum mathematically confirms the principle of PWM. The modulation process shifts the unwanted harmonic energy far away from the 50 Hz fundamental frequency, concentrating it around the 10 kHz switching frequency and its multiples.

- Current Spectrum (i_{out}): Unlike the voltage spectrum, the current spectrum shows almost no harmonic content at 10 kHz. This is due to the low-pass filtering effect of the series RL load. The impedance of the 10mH inductor is highly frequency-dependent ($Z_L = 2\pi fL$). At the fundamental frequency (50 Hz), the inductive reactance is only 3.14Ω , allowing the 50 Hz current to pass easily. However, at the switching frequency (10 kHz), the reactance increases massively to approximately 628Ω . This high impedance effectively blocks the high-frequency voltage harmonics from generating high-frequency currents, resulting in a smooth, highly sinusoidal output current waveform.

2 Part 2

This section analyzes the operation of the H-Bridge DC-AC converter using different modulation techniques. The impact of bipolar and unipolar PWM strategies on the output voltage, current, and harmonic content is evaluated.

Tasks:

2.1 Task 1:

The H-Bridge converter consists of four switches, allowing different switching states. Depending on the combination of switches, the output voltage can be +Vdc, -Vdc, or 0.

s1 (On-OFF)	s2 (On-OFF)	V_{conv} (V)
On	Off	500
Off	On	-500
On	On	0
Off	Off	0

Table 2: Switching states of s_1 and s_2 and converter voltage(V_{conv})

Data				
Name	Cursor 1	Cursor 2	Delta	Mean
Time	0.0666798	0.133334	0.0666546	
▼ s1 On s2 OFF				
Iout	50	50	0	50
▼ s1 On s2 OFF				
Vout	500	500	0	500

Figure 10: s1: ON, s2: OFF

Data					
Name		Cursor 1	Cursor 2	Delta	Mean
Time		0.0666798	0.133334	0.0666546	
▼ s1 OFF s2 On					
Iout		-50	-50	0	-50
▼ s1 OFF s2 On					
Vout		-500	-500	0	-500

Figure 11: s1: OFF, s2: ON

Data					
Name		Cursor 1	Cursor 2	Delta	Mean
Time		0.0666798	0.133334	0.0666546	
▼ s1 On s2 On					
Iout		0	0	0	0
▼ s1 On s2 On					
Vout		0	0	0	0

Figure 12: s1: ON, s2: ONN

Data					
Name		Cursor 1	Cursor 2	Delta	Mean
Time		0.0666798	0.133334	0.0666546	
▼ s1 OFF s2 OFF					
Iout		0	0	0	0
▼ s1 OFF s2 OFF					
Vout		0	0	0	0

Figure 13: s1: OFF, s2: OFF

The H-Bridge topology allows bidirectional voltage generation, which is essential for AC output synthesis.

2.2 Task 2:

The H-Bridge converter was implemented using bipolar PWM modulation. In this scheme, the output voltage switches directly between +Vdc and -Vdc depending on the comparison between the reference and carrier signals.

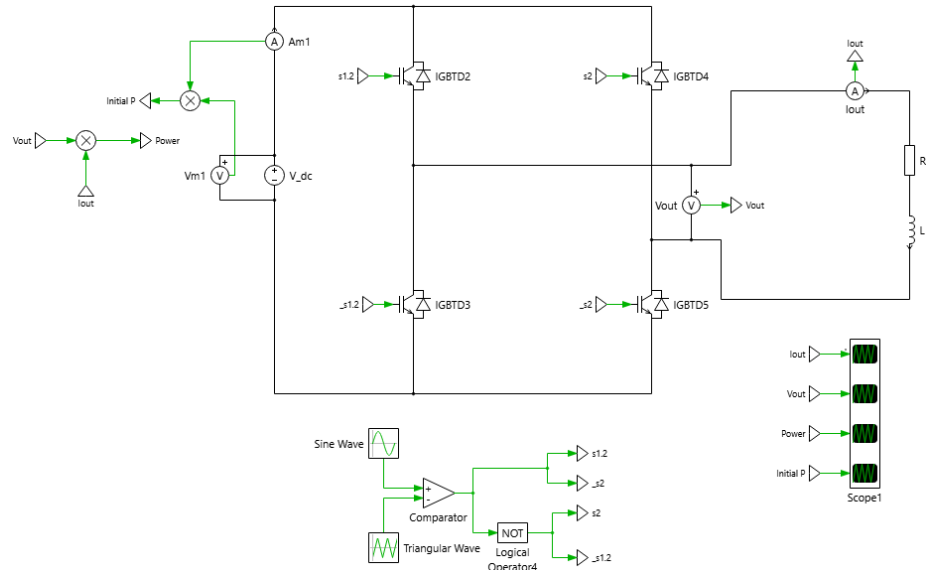


Figure 14: H-Bridge Converter Topology and Bipolar modulation scheme

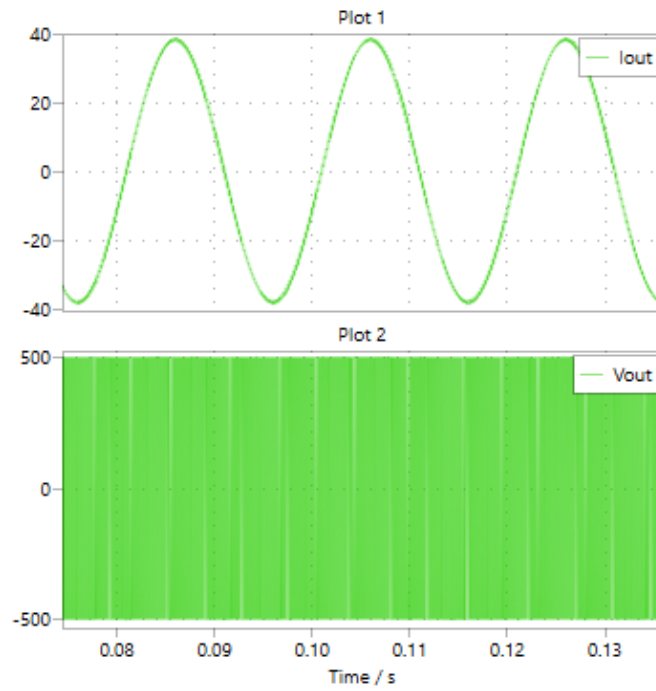


Figure 15: Plot of the output current and voltage

Bipolar modulation is simple to implement but produces higher harmonic distortion due to large voltage transitions.

2.3 Task 3:

The converter was implemented using unipolar PWM modulation. In this case, each leg of the H-Bridge is modulated separately, producing three voltage levels: $+V_{dc}$, 0, and $-V_{dc}$.

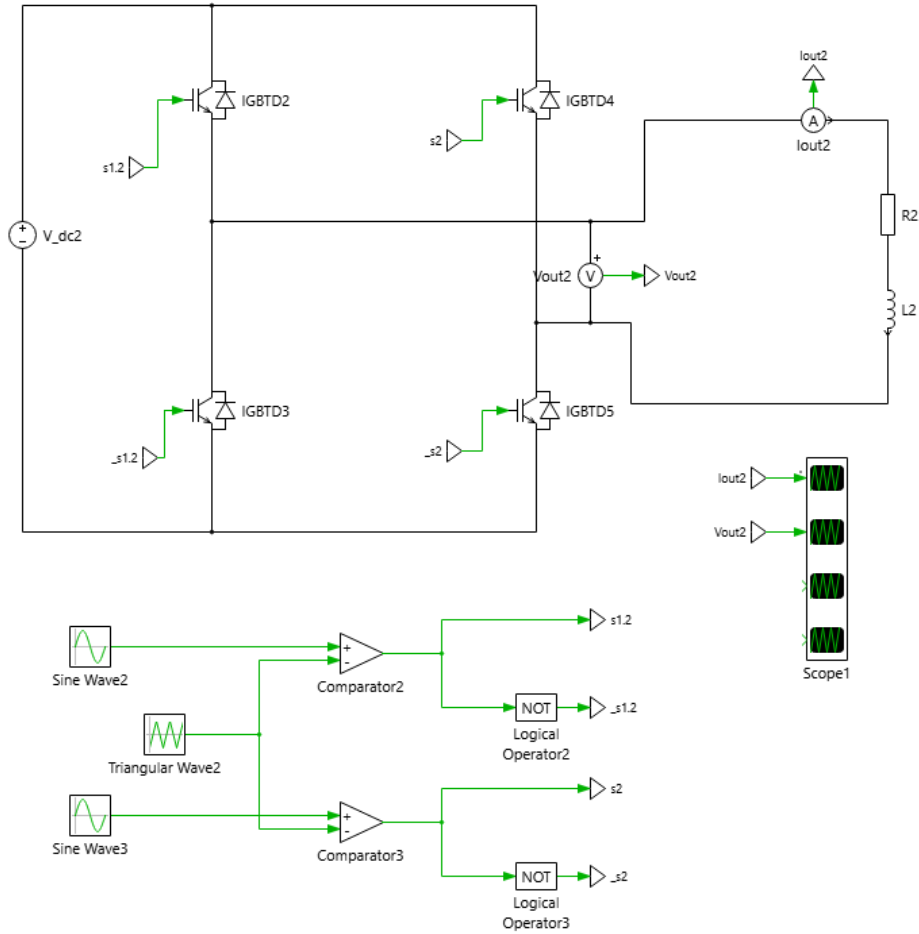


Figure 16: H-Bridge Converter Topology and Unipolar modulation scheme

Compared to bipolar modulation, the voltage transitions are smaller, reducing harmonic distortion. Unipolar modulation improves the quality of the output waveform by reducing low-order harmonics.

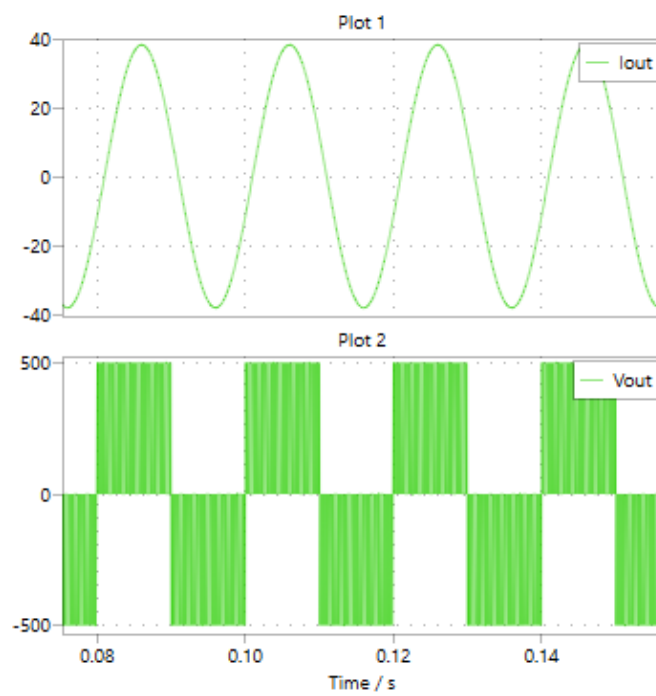


Figure 17: Plot of the output current and voltage

2.4 Task 4:

Three different modulation indexes were selected to analyze the converter performance. For each case, the fundamental component of the output voltage, current, and the delivered active power were calculated and verified through simulation.

Modulation Index (m)	V_{RMS} (V)	I_{RMS} (A)	P (W)
0.5	176.77	16.86	2840
0.8	282.84	26.98	7280
1.0	353.55	33.73	11370

Table 3: Calculated RMS voltage, current, and power for 3 modulation index

As the modulation index increases, the output voltage and current also increase, resulting in higher active power.

Modulation Index (m)	V_{RMS} (V)	I_{RMS} (A)	P (W)
0.5	278.55	16.57	2726
0.8	352.3	26.5	6979
1.0	393.9	33.15	10904

Table 4: Simulation values of RMS voltage, current, and power for each modulation index

Data					
Name		Cursor 1	Cursor 2	Delta	Mean
Time		0.0666798	0.133334	0.0666546	
Plot 1	✓				
Iout	✓	23.3946	-16.4027	-39.7973	0.58651
Plot 2	✓				
Vout	✓	500	-500	-1000	-0.106034
Plot 3	✓				
Power	✓	11697.3	8201.37	-3495.92	2726.46

Figure 18: m = 0.5

Data					
Name		Cursor 1	Cursor 2	Delta	Mean
Time		0.0666798	0.133334	0.0666546	
Plot 1	✓				
Iout	✓	37.2872	-25.9611	-63.2483	0.938392
Plot 2	✓				
Vout	✓	500	-500	-1000	-0.106034
Plot 3	✓				
Power	✓	18643.6	12980.5	-5663.05	6979.11

Figure 19: m = 0.8

Data					
Name		Cursor 1	Cursor 2	Delta	Mean
Time		0.0666798	0.133334	0.0666546	
Plot 1	✓				
Iout	✓	46.5499	-32.3335	-78.8834	1.17298
Plot 2	✓				
Vout	✓	500	-500	-1000	-0.106034
Plot 3	✓				
Power	✓	23274.9	16166.7	-7108.19	10904.7
					RMS
					33.1493
					393.915
					14805.2

Figure 20: $m = 1$

The modulation index directly controls the amplitude of the output voltage and consequently the delivered power.

2.5 Task 5:

The input active power was measured from the DC side using the instantaneous voltage and current. The output active power was obtained from the load side.

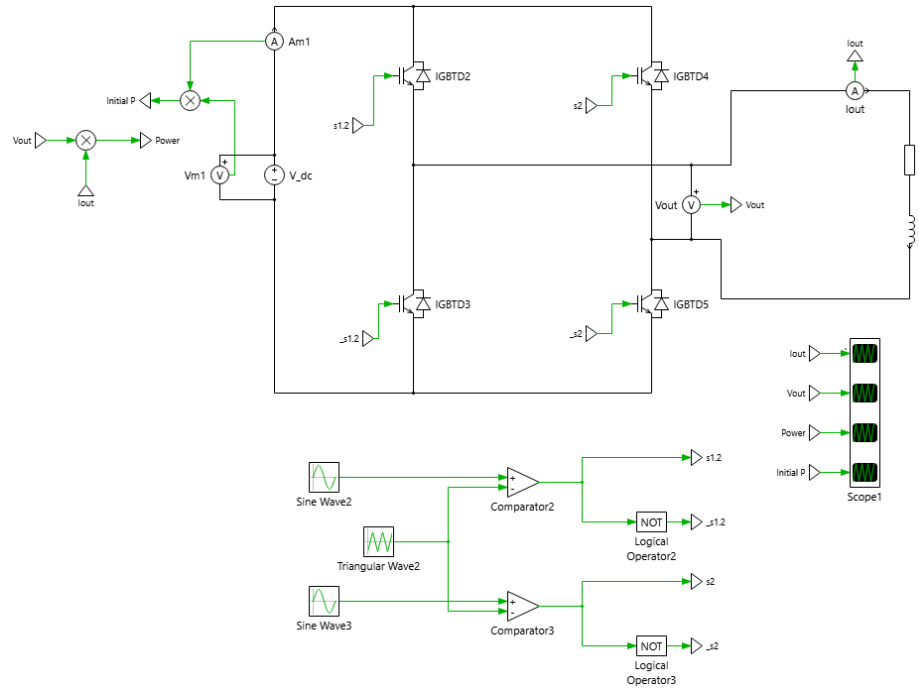


Figure 21: H-Bridge Converter Topology, Power calculation

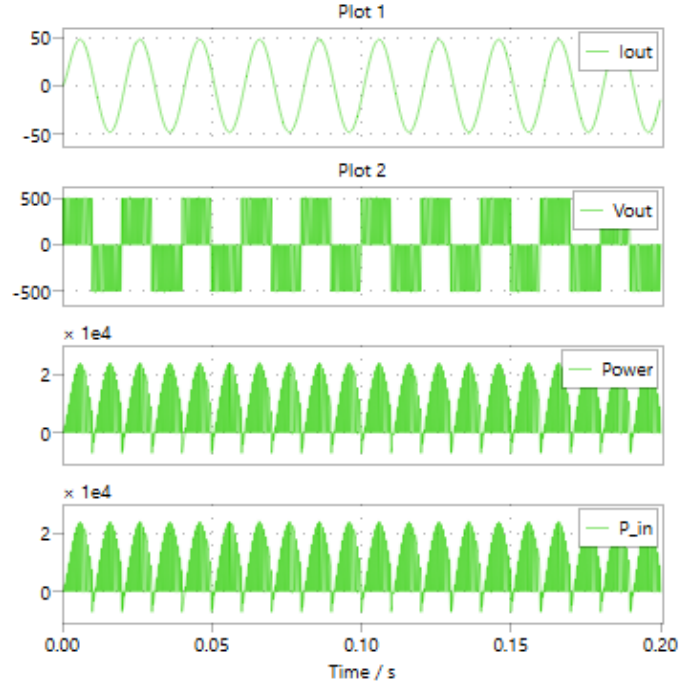


Figure 22: Plot of input Vs output active power

Comment: The input and output powers were found to be approximately equal. This confirms the energy conservation principle in the converter. Small differences may be due to numerical errors and harmonic components.

2.6 Task 6:

The frequency spectrum of the output voltage and current was obtained using FFT analysis for both bipolar and unipolar modulation.

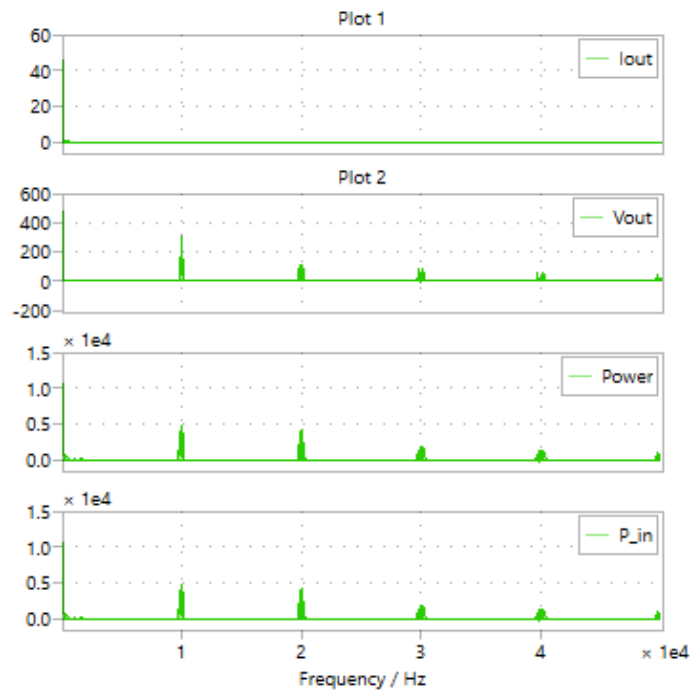


Figure 23: Bipolar Modulation Freq. Spectrum

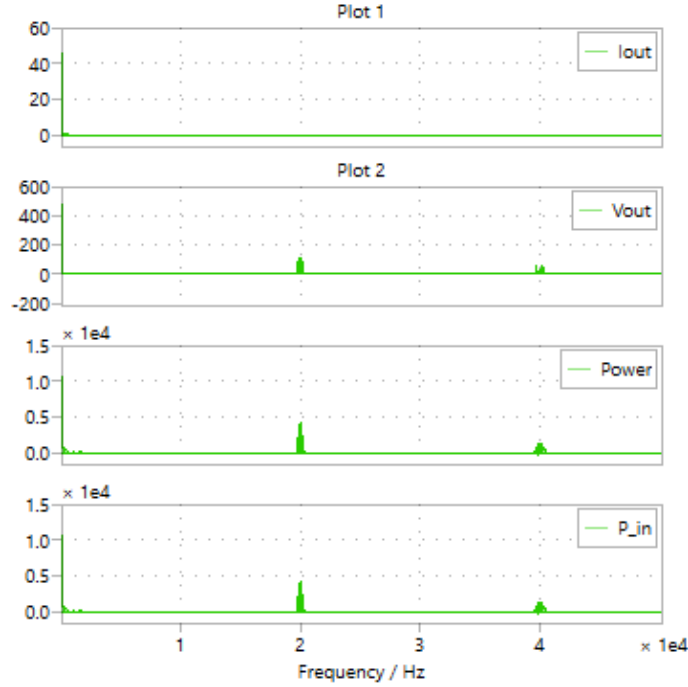


Figure 24: Unipolar Modulation Freq. Spectrum

- The fundamental frequency at 50 Hz is clearly observed in all cases.
- In bipolar modulation, the dominant harmonics appear around the switching frequency (10 kHz).
- In unipolar modulation, the dominant harmonics are shifted to higher frequencies (around 20 kHz).
- Unipolar modulation results in a cleaner output waveform with reduced low-order harmonics.
- The shift of harmonics to higher frequencies in unipolar modulation improves the overall performance of the converter.

Conclusions:

The DC-AC conversion process has been successfully analyzed using both Half-Bridge and H-Bridge topologies. The results show that both converters are capable of generating an AC output from a DC source, with the H-Bridge providing a wider range of voltage levels and improved performance. The modulation strategy plays a crucial role in the quality of the output waveform. It

was observed that bipolar modulation produces higher harmonic distortion due to larger voltage transitions, while unipolar modulation significantly improves the waveform by shifting harmonics to higher frequencies. Additionally, the modulation index directly influences the amplitude of the output voltage and the delivered power, confirming its importance in controlling the converter behavior. The comparison between input and output power demonstrates that the system follows the principle of energy conservation, with only minor differences due to non-idealities and numerical effects. Overall, the study highlights the importance of converter topology and modulation techniques in achieving efficient and high-quality DC-AC power conversion.